

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION CONTROL, MOLECULAR-WEIGHT REDUCTION, AND DOWNSTREAM FORMULATION

1. ABSTRACT

This report synthesizes six primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-rubber-derived materials. The target material is defined as NBR with reduced acrylonitrile content relative to standard grades, while retaining elastomeric performance and processability. Across the selected patents, the landscape spans three main technical directions: direct copolymer composition design, post-polymerization molecular-weight reduction, and formulation engineering for downstream use. EP3063192B1 and EP2463335B1 are the most composition-centric disclosures, describing nitrile-containing copolymers or nitrile rubber compositions with explicit acrylonitrile ranges and additional monomer/filler design intended to balance oil resistance, low-temperature flexibility, and gasoline permeation resistance. EP2145681B1 and EP2473279B1 focus on metathesis-based molecular-weight reduction of nitrile rubber, using ruthenium/osmium catalyst systems and, in one case, a boron fluoride additive system, to generate lower-MW nitrile rubbers or HNBR precursors with improved processability. EP2492306B1 addresses cross-linkable nitrile rubber compositions, emphasizing master-batch preparation of aromatic polyvalent amine crosslinkers to improve dispersion and suppress foaming during steam crosslinking. US10385647B2 is not a synthesis patent for NBR itself, but it is relevant as a formulation patent using nitrile elastomer/HNBR in swellable packers, showing how nitrile-rubber-based compositions are tuned with powder superabsorbent polymers, silica, sulfur cure systems, and brine/temperature-resistant processing windows. Overall, the evidence indicates that low-ACN NBR development is driven by careful monomer-ratio control, selective hydrogenation where needed, and downstream compounding strategies that preserve oil resistance while improving flexibility, viscosity, and application-specific durability. The most directly relevant synthesis parameters are acrylonitrile contents in the low-to-moderate range, butadiene-rich backbones, and process routes that avoid excessive degradation while enabling manageable molecular weight and cure behavior.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis degradation of nitrile rubber in the presence of a transition-metal catalyst, followed by hydrogenation to obtain lower-molecular-weight hydrogenated nitrile rubber. The process uses a nitrile rubber dissolved in monochlorobenzene, metathesized with a ruthenium catalyst and 1-hexene, then hydrogenated under high hydrogen pressure with a rhodium catalyst. - Nitrile rubber charge: 700 g — Examples 5-6 - Solvent charge: 4667 g monochlorobenzene — Examples 5-6 - Reactor volume: 10 L — Examples 5-6 - 1-Hexene addition: 4 phr — Examples 5-6 - Metathesis temperature: 22°C — Examples 5-6 - Metathesis time: 2 h — Examples 5-6 - Hydrogenation catalyst loading: 0.06 phr tris-(triphenylphosphine) rhodium chloride — Examples 5-6 - Hydrogen pressure: 85 bar — Examples 5-6 - Hydrogenation temperature: 138°C — Examples 5-6 - Hydrogenation time: 4 h — Examples 5-6 - Agitation speed: 600 rpm — Examples 5-6 - Metathesis catalyst loading: 0.04 phr — Example 1 - Metathesis catalyst loading: 0.06 phr — Example 2 - Metathesis catalyst loading: 0.08 phr — Example 3 - Metathesis catalyst loading: 0.1 phr — Example 4 - Metathesis catalyst loading: 0.07 phr — Example 5 - Metathesis catalyst loading: 0.01 phr — Example 6 - Metathesis catalyst loading: 0.1 phr — Example 7 - Solvent content: 200 ppm — Example 5 - Solvent content: 1,900 ppm — Example 6 - Solvent content: 1,300 ppm — Example 7 - Mn: 9,300 g/mol — Example 5 - Mw: 13,700 g/mol — Example 5 - PDI: 1.48 — Example 5 - Mn: 8,500 g/mol — Example 5 - Mw: 12,000 g/mol — Example 5 - PDI: 1.42 — Example 5 - Mn: 12,700 g/mol — Example 6 - Mw: 21,000 g/mol — Example 6 - PDI: 1.67 — Example 6 - Mn: 11,000 g/mol — Example 6 - Mw: 17,500 g/mol — Example 6 - PDI: 1.58 — Example 6 - Mn: 19,000 g/mol — Example 7 - Mw: 34,000 g/mol — Example 7 - PDI: 1.78 — Example 7 - Mn: 69,000 g/mol — Control #1 - Mw: 217,500 g/mol — Control #1 - PDI: 3.15 — Control #1 - Mn: 68,000 g/mol — Control #2 - Mw: 238,000 g/mol — Control #2 - PDI: 3.51 — Control #2 - Hydrogenation level: <0.9% — Examples 5-6

Relevant Experimental Evidence - Example 1: Increasing metathesis catalyst loading reduced molecular weight relative to the untreated control. - Example 2: A higher catalyst loading further reduced molecular weight and narrowed dispersity. - Example 3: Additional catalyst loading continued the trend toward lower molecular weight. - Example 4: The highest catalyst loading in the first set gave the lowest molecular weight among Examples 1-4. - Examples 5-6: Metathesis

followed by hydrogenation produced low-MW HNBR with low solvent content and low PDI. - Example 7: Changing the nitrile rubber feedstock still gave low-MW HNBR after metathesis/hydrogenation.

Technical Relevance Although not a direct low-ACN polymerization patent, this disclosure is highly relevant because it shows how nitrile rubber can be converted into lower-MW, more processable HNBR while retaining nitrile-rubber chemistry. The metathesis/hydrogenation route is important for downstream low-viscosity nitrile rubber formulations.

Patent 2

- Patent Number: EP3063192B1
- Patent Title: Copolymer rubber containing nitrile groups
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses copolymer rubbers containing nitrile groups with explicit monomer composition windows and high hydrogenation levels. The materials are described as nitrile/diene copolymers optionally containing methoxyethyl (meth)acrylate, 2-ethylhexyl (meth)acrylate, or polyethylene glycol (meth)acrylate, and the examples reference HNBR materials synthesized by emulsion polymerization of acrylonitrile and butadiene followed by optional metathesis and hydrogenation. - Acrylonitrile unit content: 10 to 60 wt% — Claim 1 - Conjugated diene unit content: 31 to 89.0 wt% — Claim 1 - Methoxyethyl (meth)acrylate / 2-ethylhexyl (meth)acrylate / polyethylene glycol (meth)acrylate content: 1 to 9 wt% — Claim 1 - Hydrogenation level of conjugated diene units: 50% or more — Claim 1 - Preferred acrylonitrile content: 20 to 50 wt% — Claim 1 - More preferred acrylonitrile content: 34 to 43 wt% — Claim 1 - Preferred diene content: 42 to 78.75 wt% — Claim 1 - More preferred diene content: 49 to 64.5 wt% — Claim 1 - Preferred termonomer content: 1.25 to 8 wt% — Claim 1 - More preferred termonomer content: 1.5 to 8 wt% — Claim 1 - Hydrogenation level: 75% or more — Claim 1 - Hydrogenation level: 85% or more — Claim 1 - HNBR A/B/C/D acrylonitrile content: 34 mol% — Table 26 - HNBR A/B/C/D RDB content: <0.9 mol% — Table 26 - HNBR A Mooney viscosity: 128 — Table 26 - HNBR B Mooney viscosity: 60 — Table 26 - HNBR C Mooney viscosity: 72 — Table 26 - HNBR D Mooney viscosity: 40 — Table 26 - HNBR A Mn: 101 kg/mol — Table 26 - HNBR B Mn: 81 kg/mol — Table 26 - HNBR C Mn: 106 kg/mol — Table 26 - HNBR D Mn: 84 kg/mol — Table 26 - HNBR A Mw: 341 kg/mol — Table 26 - HNBR B Mw: 267 kg/mol — Table 26 - HNBR C Mw: 336 kg/mol — Table 26 - HNBR D Mw: 210 kg/mol — Table 26 - HNBR A PDI: 3.40 — Table 26 - HNBR B PDI: 3.29 — Table 26 - HNBR C PDI: 3.2 — Table 26 - HNBR D PDI: 2.5 — Table 26 - Roll mill roll diameter: 80 mm — Example 1 - Roll mill roll length: 200 mm — Example 1 - Roll preheat temperature: 40°C — Example 1 - Front roll speed: 16.5 rpm — Example 1 - Rear roll speed: 20 rpm — Example 1 - Mixing time: 5 to 8

minutes — Example 1 - Vulcanization temperature: 180°C — Examples 10-13 - Vulcanization time: 20 minutes — Examples 10-13 - Open mill mixing temperature: room temperature (22°C ± unspecified) — Example 9/Section B - Vulcanization temperature: 180°C — Examples 37-44 - Vulcanization time: 15 minutes — Examples 37-44 - Internal mixer fill factor: 74% — Example 56 - Rotor speed: 60 rpm — Example 56 - Initial temperature: 60°C — Example 56 - Vulcanization temperature: 180°C — Example 56-60 - Vulcanization time: 15 minutes — Example 56-60 - Injection moulding test pressure: 50 bar — Table 41 - Injection moulding test pressure: 97 bar — Table 41 - Internal temperature: 100°C — Table 41 - Cure temperature: 190°C — Table 41 - Compound Mooney viscosity: 100 — Example 56 - Compound Mooney viscosity: 62 — Example 57 - Compound Mooney viscosity: 81 — Example 58 - Compound Mooney viscosity: 60 — Example 59 - Compound Mooney viscosity: 47 — Example 60

Relevant Experimental Evidence - Example 1: Roll-mill mixing of HNBR with carbon black, additives, and peroxide cure package was demonstrated. - Examples 10-13: Press-mixed formulations were vulcanized at 180°C for 20 minutes and showed distinct cure and tensile profiles. - Examples 37-44: Blends of inventive and non-inventive HNBR were prepared and vulcanized to compare performance. - Examples 56-60: Increasing the inventive low-viscosity HNBR reduced compound Mooney viscosity without added plasticizer. - Examples 56-60: Injection-moulding trials showed improved fill behavior at 50 and 97 bar and 190°C cure conditions.

Technical Relevance This patent is relevant because it defines nitrile-rubber compositions with explicit acrylonitrile ranges and demonstrates how hydrogenation, monomer balance, and viscosity control affect processability and performance. It provides a strong composition/process framework for low-ACN nitrile rubber and low-viscosity HNBR-derived materials.

Patent 3

- Patent Number: EP2145681B1
- Patent Title: Method for the reduction of the molecular weight of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses molecular-weight reduction of nitrile rubber by metathesis using catalyst systems that combine a metathesis catalyst with a boron fluoride adduct/additive system. The degraded nitrile rubber is then hydrogenated, and the disclosure emphasizes catalyst systems based on ruthenium or osmium complexes for nitrile-rubber depolymerization. - Hydrogenation degree of diene units: 50 to 100% — General description - Mooney viscosity of commercial HNBR: 55 to 120 — General description - Mn of commercial HNBR: about 200,000 to 700,000 — General description - PDI of commercial HNBR:

3 or greater — General description - Mooney viscosity target: less than 55 — General description - Mn target: about <200,000 g/mol — General description - Mw range after metathesis/hydrogenation: 30,000 to 250,000 — General description - Mooney viscosity range after metathesis/hydrogenation: 3 to 50 — General description - PDI after metathesis/hydrogenation: less than 2.5 — General description - Mw range after WO-A-03/002613 route: 25,000 to 200,000 g/mol — General description - Mooney viscosity after WO-A-03/002613 route: less than 25 — General description - PDI after WO-A-03/002613 route: less than 2.5 — General description - Mw range after EP-A-2 027 919 route: 11,300 to 186,000 — General description - PDI after EP-A-2 027 919 route: less than 3.1 — General description - Mw range after EP-A-2 030 988 route: 12,000 to 204,000 — General description - PDI after EP-A-2 030 988 route: less than 2.3 — General description - Mw range after EP-A-2 028 194 route: 36,000 to 240,000 — General description - PDI after EP-A-2 028 194 route: less than 2.4 — General description - Mw range after EP-A-2 027 920 route: 23,200 to 186,000 — General description - PDI after EP-A-2 027 920 route: less than 3.1 — General description - Mw range after EP-A-1 826 220 route: 10,000 to 186,000 — General description - PDI after EP-A-1 826 220 route: less than 2.4 — General description - Mw range after EP-A-1 894 946 route: 55,000 to 240,000 — General description - PDI after EP-A-1 894 946 route: less than 2.4 — General description - Mw range after US 2004/0123811 A1 route: 20,000 to 250,000 — General description - Mooney viscosity after US 2004/0123811 A1 route: 1 to 50 — General description - PDI after US 2004/0123811 A1 route: less than 2.6 — General description - Mw range after WO-A1-2005/080456 route: 20,000 to 250,000 — General description - Mooney viscosity after WO-A1-2005/080456 route: 1 to 50 — General description - PDI after WO-A1-2005/080456 route: less than 2.6 — General description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it addresses the key process bottleneck for low-viscosity nitrile rubber: reducing molecular weight without destroying nitrile-rubber microstructure. The disclosed metathesis catalyst system is directly applicable to making more processable low-ACN NBR or HNBR feedstocks.

Patent 4

- Patent Number: EP2463335B1
- Patent Title: Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubber compositions and a production process in which a latex of nitrile rubber is mixed with an aqueous dispersion of a laminar inorganic filler and then coagulated into crumbs. The nitrile rubber can

contain hydrogenated diene units and cationic monomer units, and the filler particle-size distribution is tightly controlled to improve gasoline permeation resistance. - Acrylonitrile monomer unit content: 25 to 60 wt% — Claim 1 - Conjugated diene monomer unit content: 40 to 75 wt% — Claim 1 - Cationic monomer unit content: 0 to 30 wt% — Claim 1 - Laminar inorganic filler loading: 1 to 100 parts by weight per 100 parts by weight nitrile rubber — Claim 1 - Particle diameter area ratio for particles $\geq 1.0 \mu\text{m}$: at most 30% — Claim 1 - Maximum peak particle diameter: 0.10 to 1.0 μm — Claim 1 - Frequency volume at maximum peak diameter: at least 3.0% — Claim 1 - Preferred cationic monomer content: 0.1 to 20 wt% — Claim 2 - Preferred nitrile rubber MEK-insoluble matter: 5 to 60 wt% — Claim 4

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it defines a nitrile-rubber composition with explicit acrylonitrile content and a process for improving gasoline permeation resistance through filler particle-size control. It is useful for low-ACN NBR design where composition balance and barrier performance must be maintained.

Patent 5

- Patent Number: EP2492306B1
- Patent Title: Cross-linkable nitrile rubber composition and manufacturing method for same
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a method for producing a cross-linkable nitrile rubber composition by first preparing a carboxyl group-containing nitrile copolymer rubber, then forming a master batch of a nitrile copolymer rubber and an aromatic polyvalent amine cross-linking agent, and finally blending that master batch into the carboxylated nitrile rubber. The key process feature is melting and dispersing the aromatic amine crosslinker into the master batch before final compounding to improve dispersion and suppress foaming during steam crosslinking. - Iodine value of carboxyl group-containing nitrile copolymer rubber (A): 120 or less — Claim 1 - Iodine value of nitrile copolymer rubber (b1): 120 or less — Claim 1 - Carboxyl group-containing monomer unit content: 0.1 to 20 wt% — Claim 3 - Acrylonitrile monomer unit content: 10 to 60 wt% — Claim 3 - Conjugated diene + saturated conjugated diene units: 20 to 89.9 wt% — Claim 3 - Master batch content: 0.1 to 40 parts by weight per 100 parts by weight rubber (A) — Claim 6 - Mixing ratio (b1):(b2): 20:80 to 90:10 by weight — Claim 7 - Melting point of aromatic polyvalent amine cross-linking agent (b2): 128°C — Example 1 - Master batch mixing temperature: 130°C — Example 1 - Kneading time for master batch: 5 minutes — Example 1 - Kneading time for rubber/additive premix: 5 minutes — Example 1 - Roll temperature: 50°C — Example 1 - Master batch amount: 6.77 parts — Example 1 - Cross-linking aid Noccelar DT: 2

parts — Example 1 - Carbon black FEF: 40 parts — Example 1 - Plasticizer ADK Cizer C-8: 5 parts — Example 1 - Antiaging agent Nauguard 445: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - Processing aid Phosphanol RL210: 1 part — Example 1 - Master batch B-2 amount: 10.2 parts — Example 2 - 2,2-bis[4-(4-aminophenoxy)phenyl]propane in B-2: 50 parts — Example 2 - Press cross-linked tensile strength: 26.1 MPa — Table 1, Example 1 - Press cross-linked elongation: 360% — Table 1, Example 1 - Press cross-linked hardness: 71 Duro-A — Table 1, Example 1 - Press cross-linked compression set: 15% — Table 1, Example 1 - Steam cross-linked tensile strength: 16.9 MPa — Table 1, Example 1 - Steam cross-linked elongation: 320% — Table 1, Example 1 - Steam cross-linked hardness: 64 Duro-A — Table 1, Example 1 - Foam count: 0 — Table 1, Example 1 - Press cross-linked tensile strength: 25.3 MPa — Table 1, Example 2 - Press cross-linked elongation: 380% — Table 1, Example 2 - Press cross-linked hardness: 70 Duro-A — Table 1, Example 2 - Press cross-linked compression set: 15% — Table 1, Example 2 - Steam cross-linked tensile strength: 16.5 MPa — Table 1, Example 2 - Steam cross-linked elongation: 340% — Table 1, Example 2 - Steam cross-linked hardness: 63 Duro-A — Table 1, Example 2 - Foam count: 0 — Table 1, Example 2 - Comparative Example 1 master batch amount: 3.39 parts — Table 1 - Comparative Example 2 master batch amount: 2.6 parts — Table 1 - Comparative Example 1 foam count: 10 — Table 1 - Comparative Example 2 foam count: 100 or more — Table 1

Relevant Experimental Evidence - Example 1: Master-batch preparation at 130°C enabled good crosslinker dispersion and gave strong press- and steam-cured products with no foaming. - Example 2: Reducing the amine content in the master batch still preserved good dispersion and suppressed foaming. - Comparative Example 1: Direct mixing of the aromatic amine crosslinker caused poor dispersion and inferior press-cured properties. - Comparative Example 2: Using an aliphatic polyvalent amine led to severe foaming during steam crosslinking.

Technical Relevance This patent is relevant because it shows how low-ACN or nitrile-rubber-based compositions can be made crosslinkable without sacrificing processability. The master-batch approach is especially important for practical compounding of nitrile rubbers with amine crosslinkers.

Patent 6

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions for packers that combine a nitrile elastomer with butadiene rubber, sulfur cure

package, silica, and a powder superabsorbent polymer. The nitrile elastomer may be hydrogenated, and the disclosure focuses on formulation and performance under brine, temperature, and pressure rather than on NBR polymerization itself. - Powder super absorbent polymer: potassium polyacrylate polyacrylamide copolymer — Claim 1 - Powder super absorbent polymer loading: about 125 to about 175 Phr — Claim 2 - Powder super absorbent polymer loading: about 170 to about 210 Phr — Claim 3 - Powder super absorbent polymer loading: at least about 150 Phr — Claim 4 - Powder super absorbent polymer loading: at least about 175 Phr — Claim 4 - Powder super absorbent polymer loading: at least about 200 Phr — Claim 4 - Powder super absorbent polymer loading: at least about 225 Phr — Claim 4 - Powder SAP fraction of total SAP: at least about 50% — Claim 5 - Powder SAP fraction of total SAP: at least about 60% — Claim 5 - Powder SAP fraction of total SAP: at least about 70% — Claim 5 - Powder SAP fraction of total SAP: at least about 80% — Claim 5 - Powder SAP fraction of total SAP: at least about 90% — Claim 5 - Powder SAP fraction of total SAP: at least about 95% — Claim 5 - Hydrogenation level of nitrile elastomer: about 70% to about 91% — Description - Mooney viscosity ML 1+4 (100°C): about 47 to about 72 — Description - Percent unsaturation: about 9% to about 30% — Description - Swell in 5% brine at 200°F: at least about 75% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 80% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 90% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 100% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 110% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 120% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 130% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 140% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 150% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 160% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 170% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 180% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 190% in about 96 hours — Description - Swell in 5% brine at 200°F: at least about 200% in about 96 hours — Description - Swell temperature: about 200°F — Description - Swell temperature: about 275°F — Description - Pressure resistance: at least about 3000 psi — Description - Pressure resistance: at least about 4000 psi — Description - Pressure resistance: at least about 5000 psi — Description - Pressure resistance: at least about 6000 psi — Description - Pressure resistance: at least about 70000 psi — Description - Pressure resistance: at least about 8000 psi — Description - Example 1 packer OD: 5.7 inches — Example 1 - Example 1 base pipe OD: 4.5 inches — Example 1 - Example 1 element length: 5 feet — Example 1 - Example 1 test temperature: 275°F — Example 1 - Example 1 salinity: 16% — Example 1 - Example 1 pressure requirement: 5000 psi — Example 1 - Example 1 manufacturing temperature: 330°F — Example 1 - Example 1 manufacturing pressure: 70 psi — Example 1 - Example 1 manufacturing time: 2.5 hours — Example 1 - Example 1 nylon wrap pull: 100 pounds — Example 1 - Example 1 hardness: 86 to 92 Shore A — Example 1 - Example 2 SAP loadings: 110 phr, 130 phr, 150 phr, 170 phr — Example 2 - Example 2 powder SAP mean particle size: 35.33 μm — Table 16 - Example 2 powder SAP median particle size: 5 μm — Table 16 - Example 2 medium granular SAP mean particle size: 298.67 μm — Table 16 - Example 2 medium granular SAP median particle size: 261 μm — Table 16 - Example 2 large granular SAP mean particle size: 666.23 μm — Table 16 - Example 2 large granular SAP median particle size: 652 μm — Table 16 - Example 4 HNBR content: 10% — Example 4 - Example 4

NBR content: 10% — Example 4 - Example 4 swell at 200°F, 5% brine, 96 h: 75.8% (DN3350) and 77.8% (HNBR) — Table 18 - Example 4 swell at 200°F, 16% brine, 96 h: 46.5% (DN3350) and 47.2% (HNBR) — Table 18 - Example 4 swell at 275°F, 5% brine, 96 h: 98.4% (DN3350) and 96.0% (HNBR) — Table 18 - Example 4 swell at 275°F, 16% brine, 96 h: 65.4% (DN3350) and 66.9% (HNBR) — Table 18

Relevant Experimental Evidence - Example 1: A packer with Composition B was cured at 330°F and 70 psi for 2.5 hours and held 5000 psi in 16% brine at 275°F. - Example 2: Powder, medium granular, and large granular SAPs were compared across 110 to 170 phr loadings. - Example 3: Powder SAP was predicted to reach 75% swell at about 205 phr, while granular SAPs were limited by leaching. - Example 4: A 205 phr powder-SAP composition with 10% NBR or 10% HNBR showed strong swell performance in brine. - Example 4: HNBR-based composition showed improved thermal stability and potentially lower leaching.

Technical Relevance This patent is not a synthesis route for NBR, but it is relevant because it uses nitrile elastomer/HNBR in a demanding swellable-rubber formulation. It shows how nitrile-rubber-based materials are tuned for brine resistance, swelling behavior, and high-temperature/high-pressure service.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3063192B1 and EP2463335B1 are the most composition-focused disclosures: EP3063192B1 allows 10 to 60 wt% acrylonitrile with optional acrylate termonomers, while EP2463335B1 uses 25 to 60 wt% acrylonitrile plus 0 to 30 wt% cationic monomer and laminar filler for gasoline-barrier performance.
- EP2145681B1 and EP2473279B1 both use metathesis to reduce nitrile-rubber molecular weight, but EP2473279B1 explicitly combines metathesis with hydrogenation in monochlorobenzene and reports low-Mw HNBR values, whereas EP2145681B1 emphasizes a catalyst system containing metathesis catalyst plus BF₃-type additive complexes.
- EP2473279B1 discloses the most explicit low-MW product data among the degradation patents, with Example 5 giving Mn 8,500-9,300 g/mol and Mw 12,000-13,700 g/mol, while Example 7 gives Mn 19,000 g/mol and Mw 34,000 g/mol.
- EP2492306B1 is distinct from the others because it is a compounding/crosslinking patent rather than a polymerization patent; its key innovation is master-batch dispersion of aromatic polyvalent amine crosslinkers at 130°C to suppress foaming.
- US10385647B2 is formulation-driven and uses nitrile elastomer/HNBR as a swellable-packers base, with powder SAP loadings of 125 to 225 phr and service conditions of 200°F to 275°F and up to 5000 psi, showing downstream application requirements rather than synthesis conditions.
- Across the set, acrylonitrile content windows remain broad in the composition patents, but the process patents focus on lowering molecular weight and viscosity to improve processability

without sacrificing nitrile-rubber performance.

4. CONCLUSION

The patent evidence shows that low-acrylonitrile NBR development is best approached through controlled copolymer composition, followed by process or formulation steps that preserve elastomeric performance while improving processability and end-use durability. The most directly useful synthesis windows are acrylonitrile contents in the low-to-moderate range, especially 10 to 60 wt% depending on the application, with butadiene-rich backbones and, where needed, selective hydrogenation. For lower viscosity and better handling, metathesis-based molecular-weight reduction is strongly supported, particularly when paired with hydrogenation and solvent systems such as monochlorobenzene. For practical compounding, master-batch dispersion of crosslinkers and carefully selected filler/SAP systems are important to avoid foaming, leaching, and poor cure behavior. Based on the evidence, a recommended low-ACN NBR strategy is to target acrylonitrile in the lower end of the disclosed windows, maintain butadiene as the major comonomer, and use controlled post-polymerization molecular-weight reduction only as needed to reach processable Mooney viscosity while preserving oil resistance and cure response.

5. REFERENCES

- EP2473279B1 | Metathesis of a nitrile rubber in the presence of a transition metal catalyst | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473279B1/en>
- EP3063192B1 | Copolymer rubber containing nitrile groups | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP3063192B1/en>
- EP2145681B1 | Method for the reduction of the molecular weight of nitrile rubber | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2145681B1/en>
- EP2463335B1 | Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ... | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2463335B1/en>
- EP2492306B1 | Cross-linkable nitrile rubber composition and manufacturing method for same | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2492306B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>